HONGYI XIE

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Fudan University, Shanghai, China

Google Scholar: https://scholar.google.com/citations?user=uEO5wRMAAAAJ

EDUCATION

Fudan University (QS 39) *Sep 2022 - present (expected to graduate in Jun 2026)*

Ph.D., Environmental Management, Supervisor: Prof. Yutao Wang

National University of Singapore (QS 8) *Aug 2024 - Aug 2025*

Joint Ph.D., Civil and Environmental Engineering, Supervisor: Prof. Yang Guo

Beijing Normal University (QS 271) *Sep 2017 - Jun 2020*

M.S., with honors, Environmental Science, Supervisor: Prof. Zhifeng Yang and Prof. Yujun Yi

Zhejiang University (QS 47) *Sep 2013 - Jun 2017*

B.S., Environmental Science

EMPLOYMENT

Engineer, Transport Planning and Research Institute, Ministry of Transport *Jul 2020 - Jul 2022*

Research assistant, National Taiwan University (QS 68) *Jul 2015 - Sep 2015*

RESEARCH INTEREST

Co-benefits of cross-sector climate policies; Constraints of renewable energy scale-up; Sustainable low-carbon technology supply chains

PUBLICATIONS

In revision and under review:

1. **Xie, H.**, Wu, Y., Li, Z., Wang, Y., Guo, Y. Addressing water resource constraints for electrolytic hydrogen demand in China. (Minor revision for ***Nature Sustainability***)
2. **Xie, H.**, Li, H., Wang, S., Han, Z., Chen, B., Lv, H., Wang, Y. Shifting retired battery disposal from metallurgical recycling to high-value utilization for optimizing decarbonization in China. (Major revision for ***Environmental Science and Ecotechnology***)
3. Xie, G., Bai, Y., Wang, Q., **Xie, H.* (corresponding author)**. Rethinking Urbanization under the Socio－Economic－Ecosystem Nexus: Insights from an Emergy Perspective in Shandong Province, China. (Under review for ***Environment, Development and Sustainability***)
4. Li, H., Chen, B., **Xie, H.**, Yu, H., Dai, M., Xue, M., Long, X., Wang, J., Han, Z., Sun, M., Wang, J., Guo, Y., Lin, J., Wang, Y. The co-benefits of the use of fly ash as a clinker substitute will be undermined by coal phaseout. (Minor revision for ***the Innovation***)

Peer reviewed:

1. **Xie, H.**, Chen, B., Wang, C., Xie, W., Long, X., Zhang, D., Dai, M., Han, Z., Song, Y., Li, J., Wang, Y. (2025). Uncertain cross-sector climate actions could undermine air pollution reduction co-benefits in China's power and passenger car sectors. ***Engineering***, In press. https://doi.org/10.1016/j.eng.2025.09.006

2. **Xie, H.**, Chen, B., Dai, M., Han, Z., Bai, Y., Wang, Y. (2024). Upgrading passenger vehicle emission standard help to reduce China's air pollution risk from uncertainty in electrification. ***Environmental Science & Technology***, 58, 5325-5335. (Supplementary cover paper). https://doi.org/10.1021/acs.est.3c10078

3. **Xie, H.**, Li,Y., Wang, C. (2025). Decarbonizing inland waterway transport: scenario analysis and mitigation strategy for Jiangxi Province, 2019–2050. ***ENGINEERING Environment***, 19(8), 113. https://doi.org/10.1007/s11783-025-2033-4

4. Long, X., Chen, B., Dai, M., **Xie, H.**, Chen, P., Meng, J., Wang, Y. (2025). Ambitious climate targets can make the phaseout of India's coal-fired power plants cost-beneficial. ***Nature Communications***, In press. https://doi.org/10.1038/s41467-025-66580-4

5. Shui, B., Cai, Z., **Xie, H.**, Liu, F., Luo, X. (2026). Harnessing Policy Synergies to Decarbonize China's Light-Duty Passenger Vehicle Fleet. ***Journal of Cleaner Production***, 538, 147236. https://doi.org/10.1016/j.jclepro.2025.147236

6. Zhang, Y., Dai, M., Chen, B., **Xie, H.**, Li, Z., Wang, Y. (2025). A Product-based Life-cycle Integration Platform for sustainable plastic bottles from agricultural biowaste. ***Chemical Engineering Journal***, 524, 169320. https://doi.org/10.1016/j.cej.2025.169320

7. Bai, Y., Li, H., Chen, B., **Xie, H.**, Wang, Y. (2025). Managing nitrogen metabolism of animal husbandry and aquaculture could mitigate nitrogen threat in the Main Cities of Yellow River Delta. ***Cleaner Environmental Systems***, 17,100273. https://doi.org/10.1016/j.cesys.2025.100273

8. Wang, C., Kong, Y., Lu, X., **Xie, H.**, Teng, Y., Zhan, J. (2024). Rethinking Regional High-Quality Development Pathways from a Carbon Emission Efficiency Perspective. ***Land***, 13(9), 1441. https://doi.org/10.3390/land13091441

9. Li, H., Bian, Y., Liu, M., Lin, J., Dai, M., **Xie, H.**, Yu, H., Chen, B., Xue, M., Li, Z., Yin, J., Xue, L. CO_2 mineralization and utilization by tailings sand in China for potential carbon sinks and spatial project layout. ***Resources, Conservation and Recycling***, 206, 107598. https://doi.org/10.1016/j.resconrec.2024.107598

10. Xie, W., Yang, X., Han, Z., Sun, M., Li, Y., **Xie, H.**, Yu, H., Chen, B., Fath, B., Wang, Y. (2024). Urban sector land use metabolism reveals inequalities across cities and inverse virtual land flows. ***Resources, Conservation and Recycling***, 202, 107394. https://doi.org/10.1016/j.resconrec.2023.107394

11. Fang, Y., **Xie, H.**, Chen, B., Han, Z., An, D., Cai, W., Zhang, W., Wang, Y. (2024). Achieving carbon neutrality in Shanghai's municipal wastewater treatment sector requires coordinated water conservation and technical improvement. ***Journal of Cleaner Production***, 443, 141134. https://doi.org/10.1016/j.jclepro.2024.141134

12. Han, Z., Xie, W., Yu, H., **Xie, H.**, Li, Y., Wang, Y. (2024). Rethinking industrial land-use in American rust cities towards sustainability based on a block-level model. ***Journal of Environmental Management***, 352, 120067. https://doi.org/10.1016/j.jenvman.2024.120067

13. Dai, M., Sun, M., Chen, B., **Xie, H.**, Zhang, D., Han, Z., Yang, L., Wang, Y. (2023). Advancing sustainability in China's pulp and paper industry requires coordinated raw material supply and waste paper management. ***Resources, Conservation and Recycling***, 198, 107162. https://doi.org/10.1016/j.resconrec.2023.107162

14. Gao, Y., Yi, Y., Chen, K., **Xie, H.** (2023). Simulation of suitable habitats for typical vegetation in the Yellow River Estuary based on complex hydrodynamic processes. ***Ecological Indicators***, 154, 110623. https://doi.org/10.1016/j.ecolind.2023.110623

15. Liang, Z., Deng, H., **Xie, H.**, Chen, B., Sun, M., Wang, Y. (2023). Rethinking the paper product carbon footprint accounting standard from a life-cycle perspective. ***Journal of Cleaner Production***, 393, 136352. https://doi.org/10.1016/j.jclepro.2023.136352

16. **Xie, H.**, Li,Y. (2022). An ecological water replenishment model of urban lake riparian plant restoration based on the groundwater – vegetation interactions. ***Ecological Engineering***, 176, 106510. https://doi.org/10.1016/j.ecoleng.2021.106510

17. **Xie, H.**, Yi, Y, Hou, C., Yang, Z. (2020). In situ experiment on groundwater control of the ecological zonation of salt marsh macrophytes in an estuarine area. ***Journal of Hydrology***, 585, 124844. https://doi.org/10.1016/j.jhydrol.2020.124844

18. Yi, Y., **Xie, H.**, Yang, Y., Zhou, Y., Yang, Z. (2020). Suitable habitat mathematical model of common reed (Phragmites australis) in shallow lakes with coupling cellular automaton and modified logistic function. ***Ecological Modelling***, 419, 108938. https://doi.org/10.1016/j.ecolmodel.2020.108938

19. Yi, Y., **Xie, H.**, Song, J., Yang, Z. (2021). Simulation of salt marsh vegetation community's suitable habitat in Yellow River Estuary I: theory. ***Journal of Hydraulic Engineering (水利学报)***, 52(03), 255-264. (In Chinese). https://doi.org/10.13243/j.cnki.slxb.20200542

20. Yi, Y., **Xie, H.**, Song, J., Yang, Z. (2021). Simulation of salt marsh vegetation community's suitable habitat in Yellow River Estuary Ⅱ:application. ***Journal of Hydraulic Engineering (水利学报)***, 52 (04), 401-408. (In Chinese). https://doi.org/10.13243/j.cnki.slxb.20200544

21. **Xie, H.**, Liu, J., Gao, Y., Li, Q., Gao, J., Yang, S., Zhu, G. (2022). Method and Application of Regional Highway Resources and Environment Carrying Capacity Evaluation. ***Highway (公路)***, 67 (09), 407-412. (In Chinese).

22. Wang, C., **Xie, H.**, Zhang, X., Zeng, X. (2025). Spatiotemporal variations of carbon source-sink matching in the Beijing-Tianjin-Hebei urban agglomeration. ***Chinese Journal of Ecology (生态学杂志)***, 44 (04), 1343-1354. (In Chinese). https://doi.org/10.13292/j.1000-4890.202504.041

23. Wang, S., Chen, B., Dai, M., **Xie, H.**, Zhang, D., Sun, M., Wang, Y. (2025). Carbon footprint accounting and impact factor analysis of China's bioeconomy. ***Environmental Engineering (环境工程)***,43(8), 233-243. (In Chinese). https://doi.org/10.13205/j.hjgc.202508022

Reports, books and standards:

1. United Nations Environment Programme (2024). Global Bioeconomy Assessment: Coordinated Efforts of Policy, Innovation, and Sustainability for a Greener Future. https://wedocs.unep.org/20.500.11822/45332

2. China Environment Publishing Group (2020). Theory and Practice of Risk Analysis and Emergency Capacity Planning for Oil Spills on Water. ISBN: 978-7-0000-0000-0

3. Chinese Group Standard of China Institute of Navigation (2023). Guidelines for the construction planning of ship oil spill response capacity of inland water (T/CIN 025-2023). https://www.cinnet.cn/zh-hans/system/files/files/25.nei_he_chuan_bo_yi_you_ying_ji_neng_li_jian_she_gui_hua_bian_zhi_zhi_nan_.pdf

Awards

Joint PhD Training Program for State-constructed High-level Universities, China Scholarship Council (2024)

Second-class Academic Scholarship, Fudan University (2024)

Third Prize, Excellent Consulting Achievement Award in Port and Waterway Engineering, China Waterborne Transport Construction Association (2023)

Outstanding Graduate, Beijing Normal University (2020)

First-/Second-class Academic Scholarship, Beijing Normal University (2018, 2019)

Conferences & Presentations

2025 – 12th International Conference on Industrial Ecology, Singapore – Oral Presentation

2024 – Nature Conference on Air Pollution and Climate Change, Beijing – Poster Presentation

2023 – International Conference on Resource Sustainability, United Kingdom – Oral Presentation

2018 – 1st International Student Forum on Environmental Ecology, Beijing – Poster Presentation

Technical Skills

Analysis: Techno-economic assessment; Life Cycle Assessment; Prospective LCA (premise); Material Flow Analysis; Emission inventory; Spatial Analysis (ArcGIS)

Modeling: Power system model (GCAM); Multi-Agent modeling (AnyLogic); Hydrodynamic modeling (MIKE 21, Visual MODFLOW)

Coding: Matlab, R, Python

Service

Journal referee: *Nature Cities*; *Environmental Science & Technology*; *Resources, Conservation and Recycling*; *Journal of Cleaner Production*; *Journal of Environmental Management*; *Ecological Indicators*; *Geography and Sustainability; iScience*; *Ecological Economics*; *Journal of Hydrology Regional Studies*; *Regional Studies in Marine Science*; *Journal of Sea Research*; *Heliyon*; *Energy, Ecology and Environment*; *Results in Engineering; Unconventional Resources*

Referees

Prof. Yutao Wang (Doctoral Supervisor), Fudan University, yutaowang@fudan.edu.cn

Prof. Yang Guo (co-Supervisor), Beijing Normal University, guoyang@bnu.edu.cn